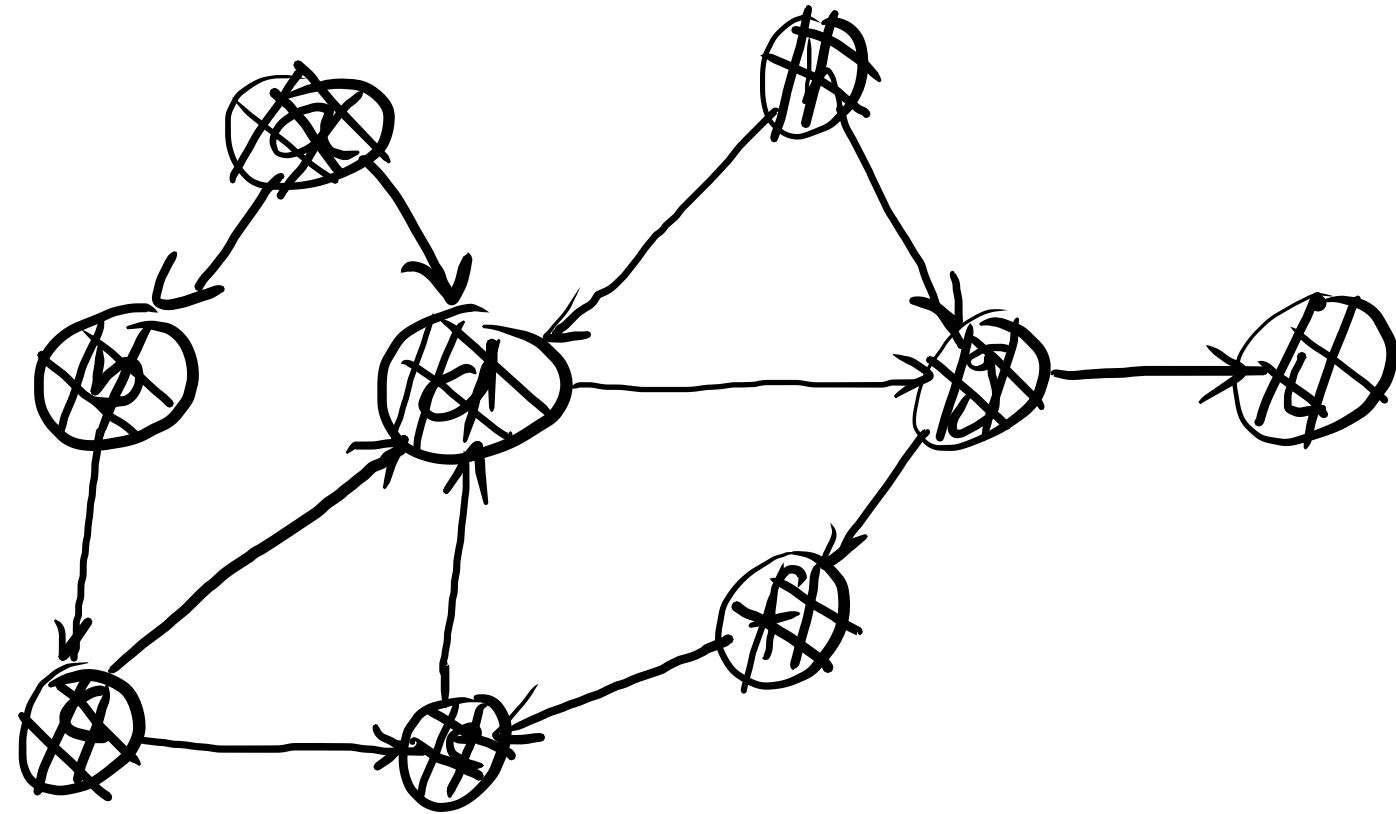


DFS Example:



S

1	2	3	4	5	6	7	8
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1	2	3	4	5	6	7	8
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1	2	3	4	5	6	7	8
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DFS Forest

d
|
g
|
f
|
e
T₁

|
i

a
|
b
|
c
T₂

h
T₃

DFS(G)

for each vertex $u \in G.V$
 $u.color = WHITE$
 $u.\pi = Nil$

for each vertex $u \in G.V$
 if $u.color == WHITE$
 DFS-rec(G, u)

DFS-rec(G, u)

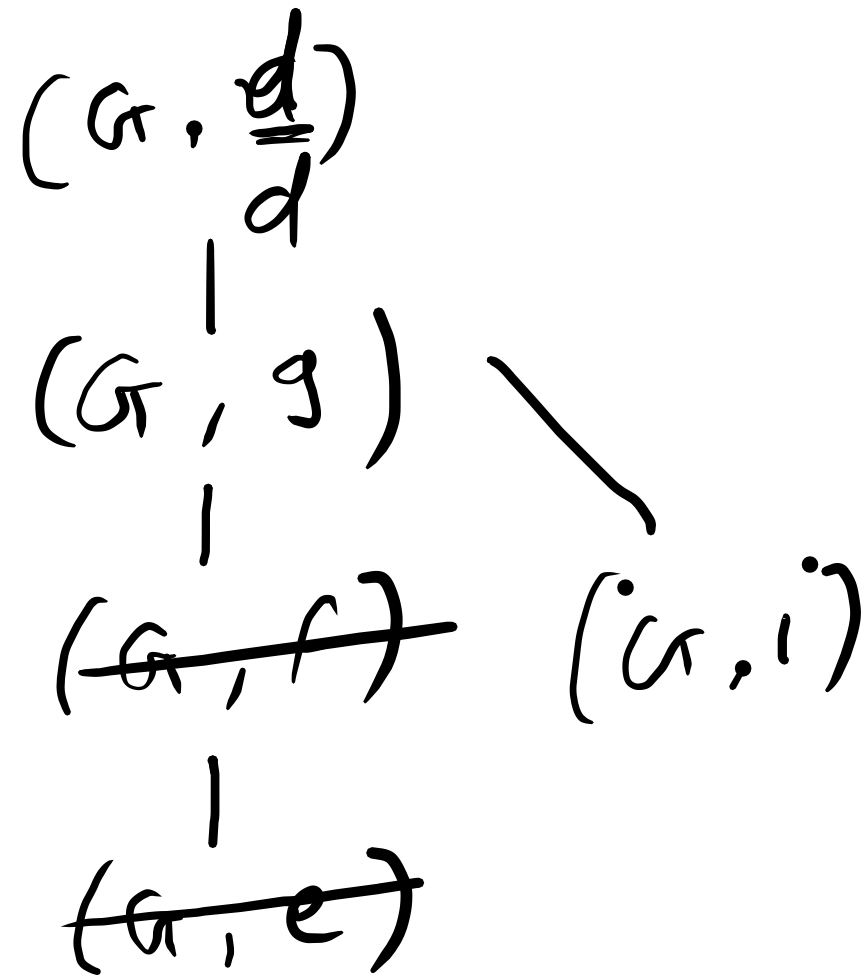
$u.color = GRAY$

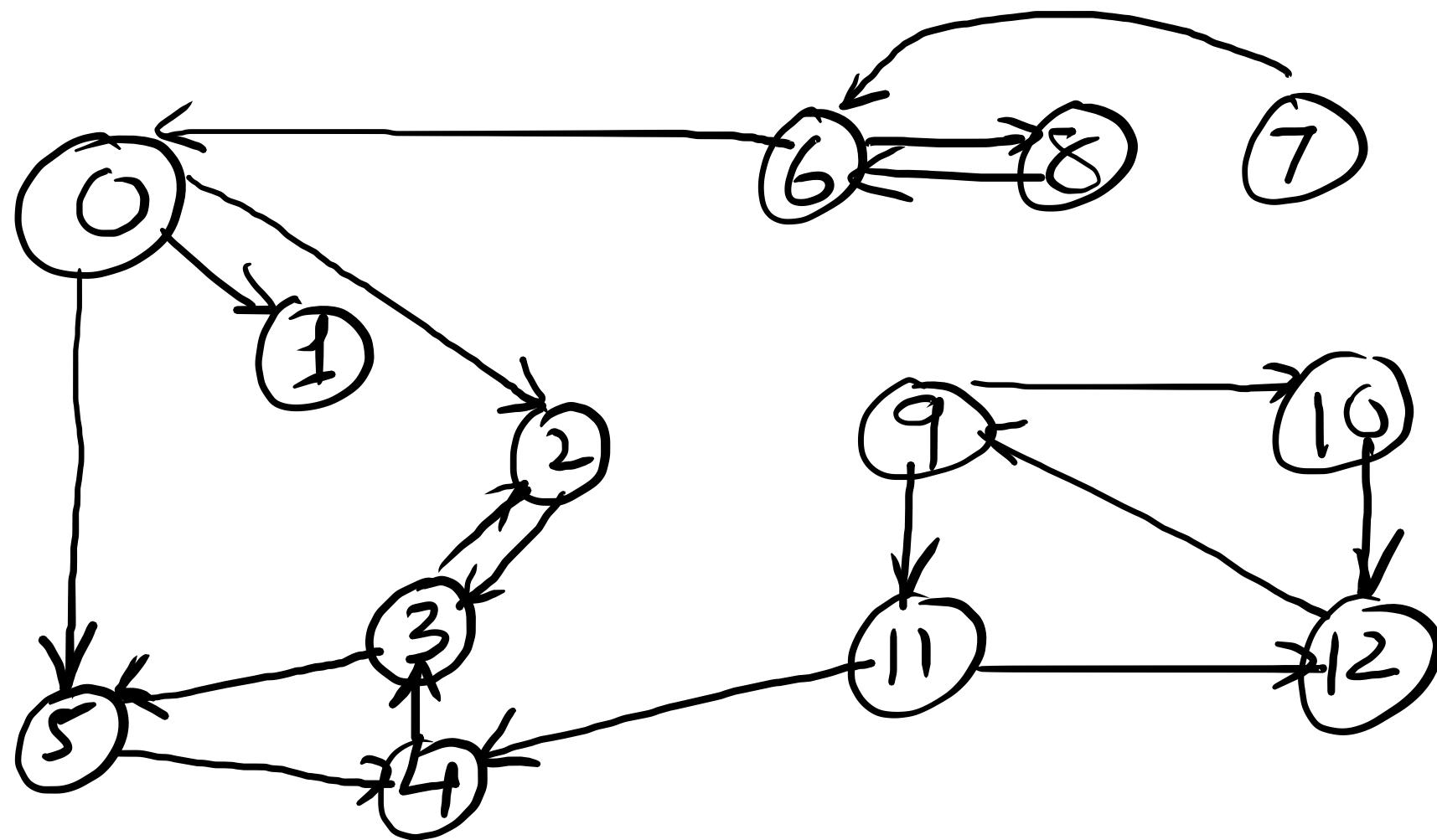
 for each vertex $v \in adj[u]$
 if $v.color == WHITE$

$v.\pi = u$

 DFS-rec(G, v)

$u.color = BLACK$





BFS → "spreading virus"
DFS → "Exploring web one click at a time"

Applications of DFS: Detecting Unreachable code

if age > 18 goto a;

if age > 21 goto b;
goto c;

d: _____

a: _____

b: _____

c: _____

for (i = 0; i < 10; i++)
a = a + 1

MOV R1, #0 → i

label2: CMP R1, #10

JGE label1

LD R2, <1000> → a

ADD $\frac{R2}{D}, \frac{R2}{S1}, \frac{\#1}{S2}$

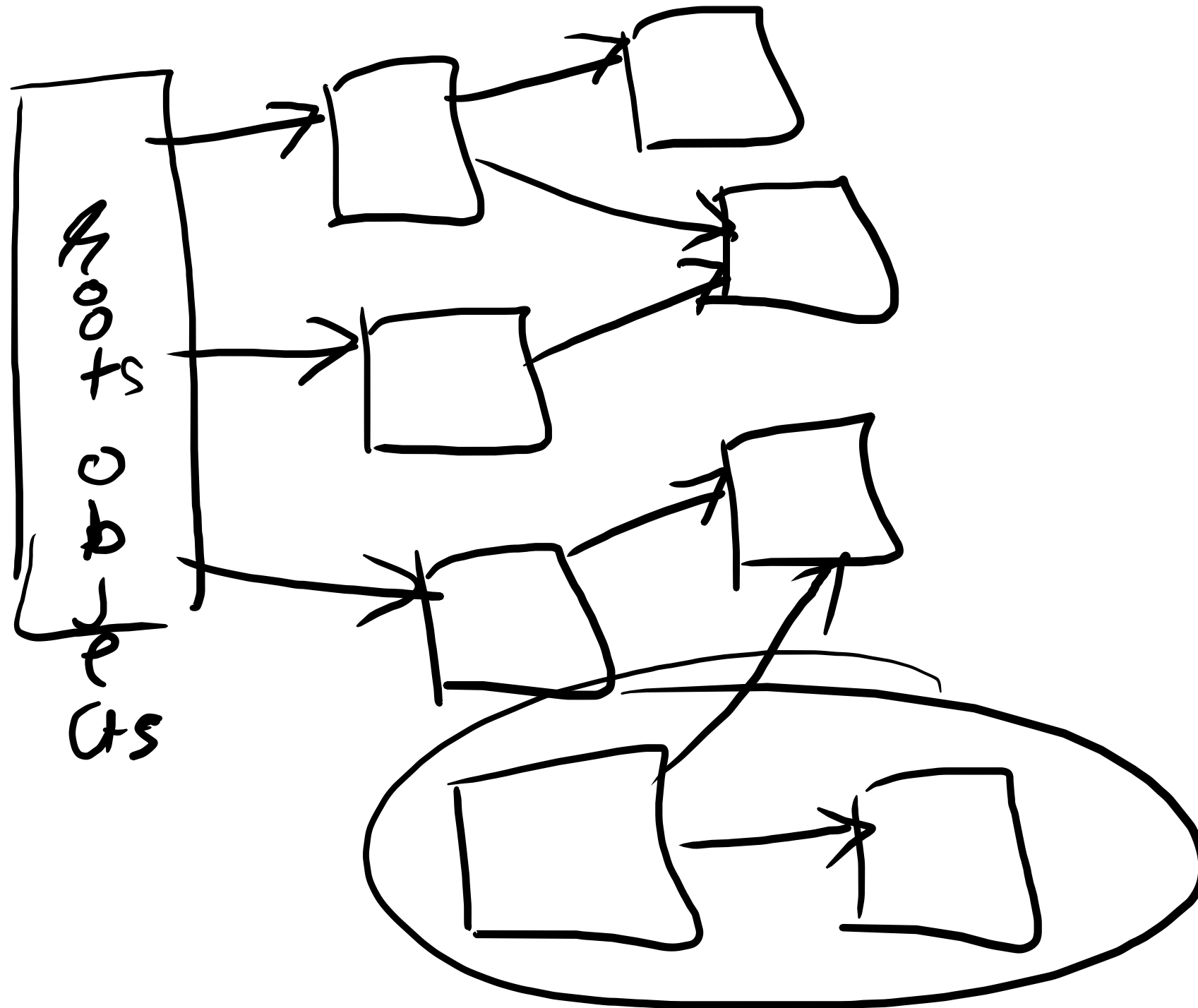
ADD R1, R1, 1

JMP label2

label1: _____

Garbage collection:

while(1)
malloc()



$$\textcircled{a f(n) \over b}$$

-

$$\textcircled{a f\left(\frac{n}{b}\right)}$$

$n = \text{number.}$

~~$$b = \lg n$$~~

~~$$O(b)$$~~

$$O(\lg n)$$

~~$$n \geq 2$$~~

$$\underline{\lg n}$$

←

